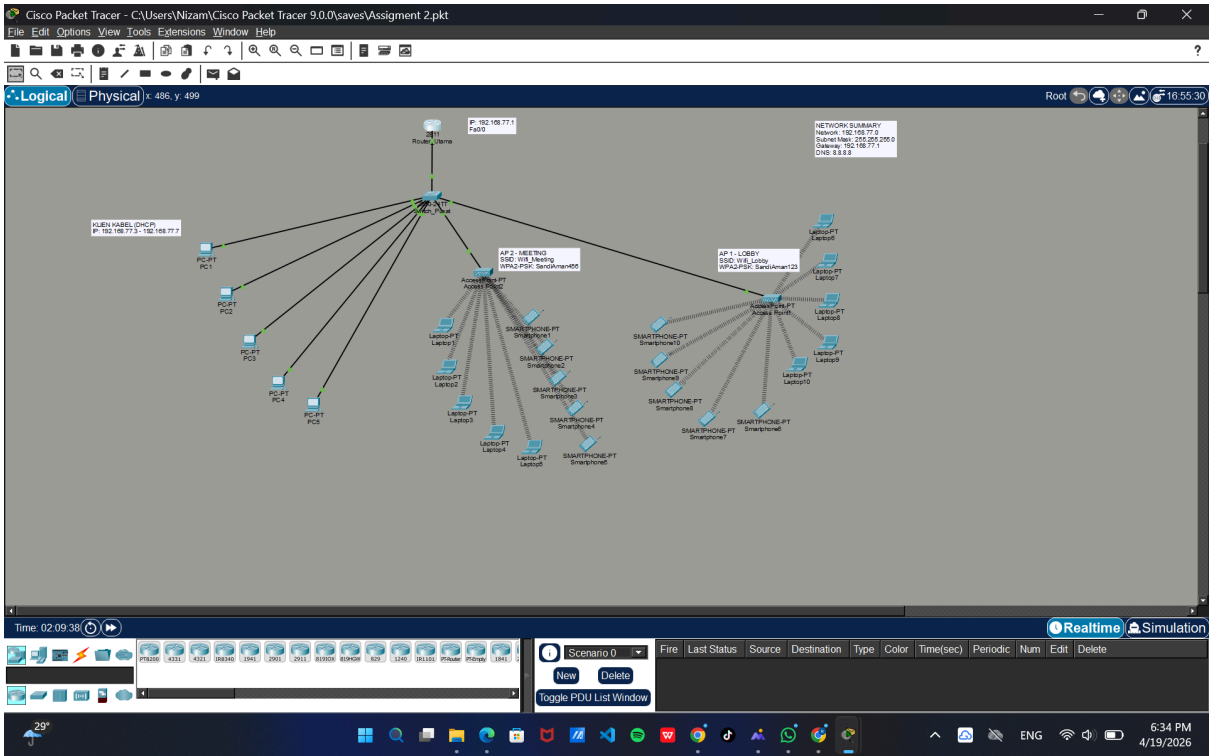


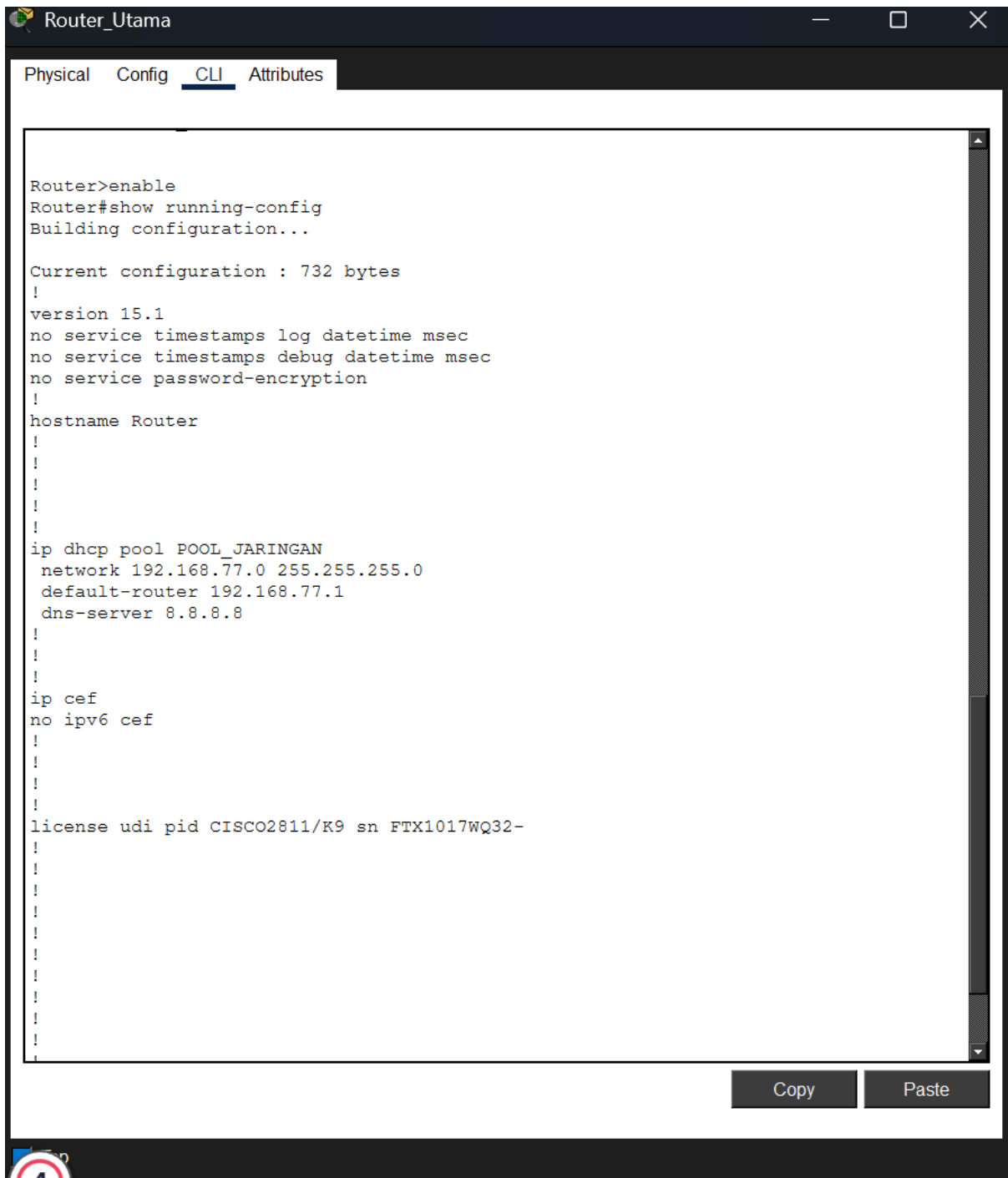
ASSIGNMENT COMPUTER NETWORK

Name: Nizamuddin Giffa Abdul Aziz Zahrani

Class: IS ERP 2

ID: 012202400062





Access Point1

Physical

Config

Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status

On

SSID

Wifi_Lobby

2.4 GHz Channel

6

Coverage Range (meters)

140.01

Authentication

Disabled

WPA-PSK

WEP

WPA2-PSK

WEP Key

PSK Pass Phrase

SandiAman123

User ID

Password

Encryption Type

AES

Top

Device	SSID (Wi-Fi Name)	Security Mode	Password (PSK)	Subnet / IP Range

Access Point 1	Wifi_Lobby	WPA2-PSK (AES)	SandiAman123	192.168.77.0/24
Access Point 2	Wifi_Meeting	WPA2-PSK (AES)	SandiAman456	192.168.77.0/24

Security: All wireless devices are configured using the **WPA2-PSK (AES)** encryption standard to ensure a secure and encrypted connection across the network.

IP Management: IP Address allocation for all clients (Laptops & Smartphones) is centrally managed by the Main Router (Router_Utama) via the **DHCP Pool** service, which effectively minimizes the risk of IP conflicts.

Access Point2

PhysicalConfigAttributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status

SSID

2.4 GHz Channel

Coverage Range (meters)

Authentication

Encryption Type

☒ On

Wifi_Meeting

6

140.00

☒ Disabled

☒ WPA-PSK

☒ WEP

☒ WPA2-PSK

WEP Key

PSK Pass Phrase

User ID

Password

SandiAman456

AES

Top

Laptop1

PhysicalConfigDesktopProgrammingAttributes

MODULES

WPC300N

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

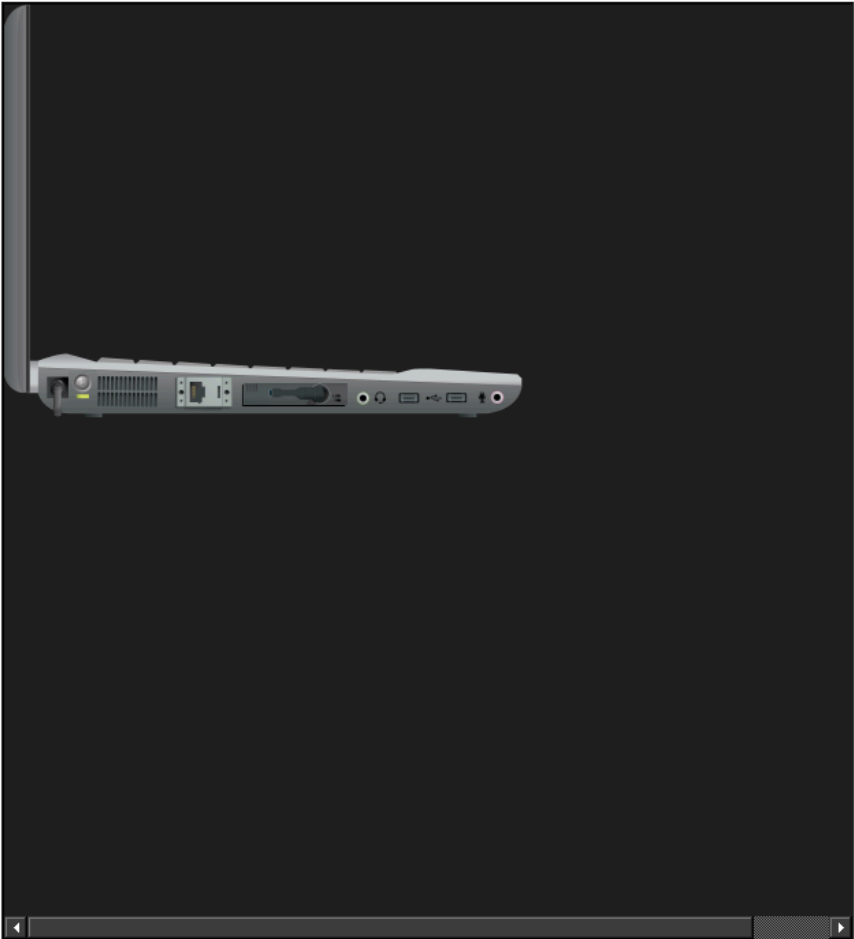
PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

PT-HEADPHONE

PT-MICROPHONE

Zoom InOriginal SizeZoom Out



Customize Icon in Physical View

Customize Icon in Logical View

The PT-LAPTOP-NM-1W-A module provides one 5GHz wireless interface suitable for connection to wireless 802.11a networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop2

Physical

Config

Desktop

Programming

Attributes

MODULES

WPC300N

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

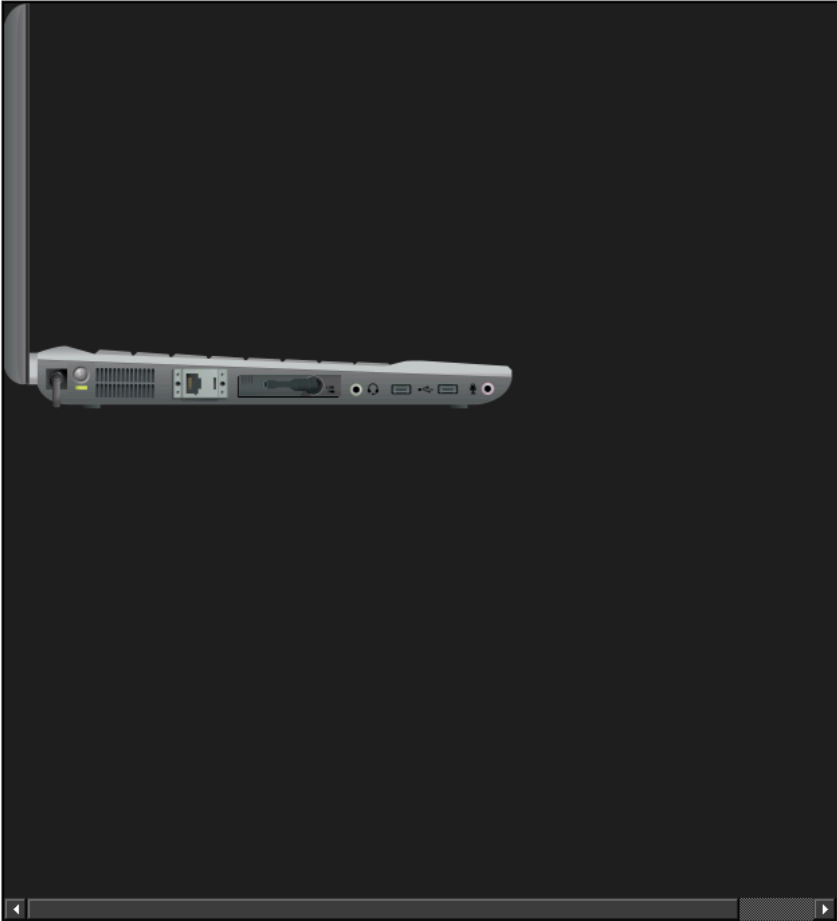
PT-HEADPHONE

PT-MICROPHONE

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View

Customize Icon in Logical View

The PT-LAPTOP-NM-1W-A module provides one 5GHz wireless interface suitable for connection to wireless 802.11a networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop3

Physical

Config

Desktop

Programming

Attributes

MODULES

WPC300N

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

PT-HEADPHONE

PT-MICROPHONE

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View

Customize Icon in Logical View

The PT-LAPTOP-NM-1W module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop4

Physical

Config

Desktop

Programming

Attributes

Clo

MODULES

WPC300N

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

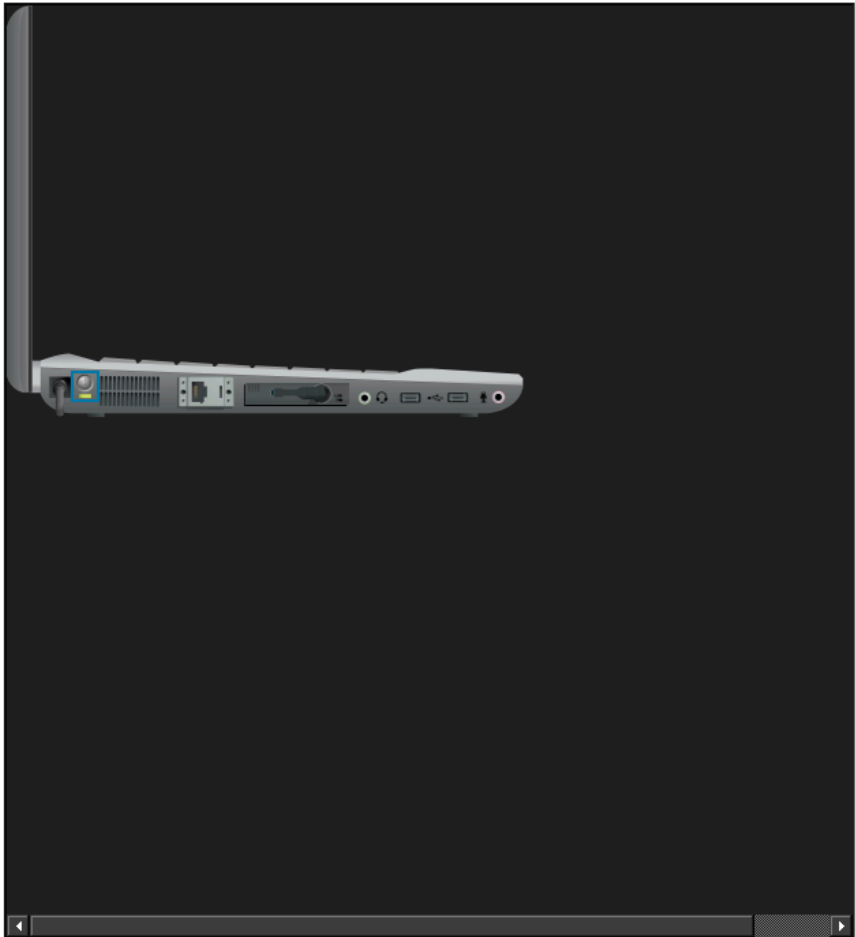
PT-HEADPHONE

PT-MICROPHONE

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View

Customize Icon in Logical View

The PT-LAPTOP-NM-1W module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop5

Physical

Config

Desktop

Programming

Attributes

MODULES

WPC300N

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

PT-HEADPHONE

PT-MICROPHONE

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View

Customize Icon in Logical View





The PT-LAPTOP-NM-1W module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop6

Physical

Config

Desktop

Programming

Attributes

MODULES

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

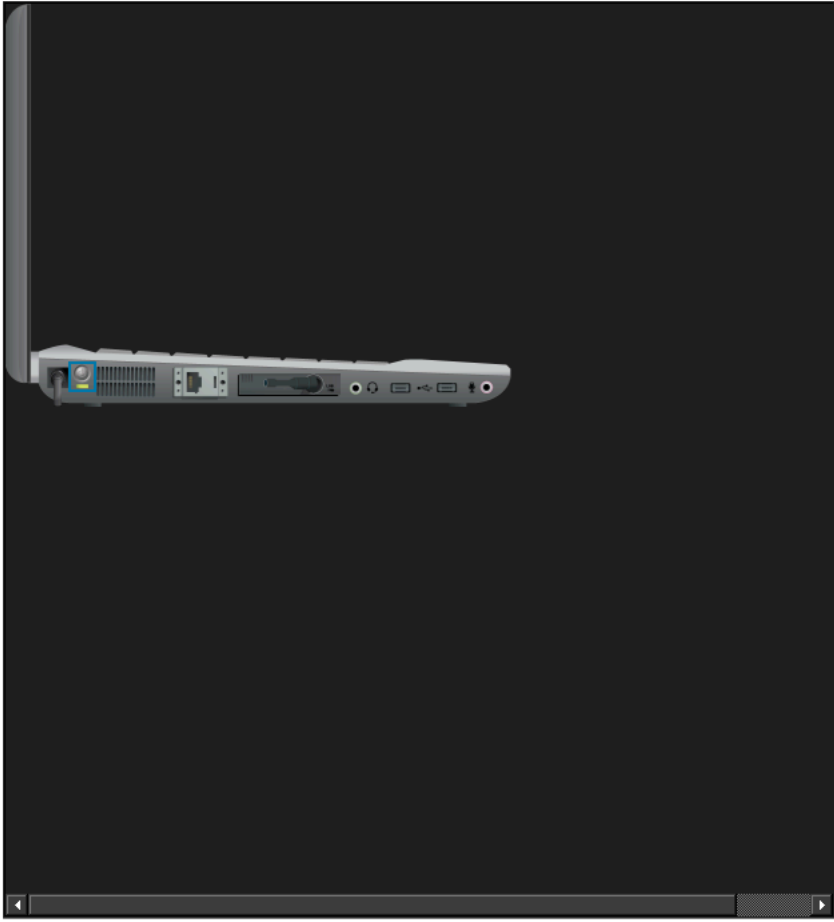
PT-HEADPHONE

PT-MICROPHONE

Zoom In

Original Size

Zoom Out



Customize
Icon in
Physical View



Customize
Icon in
Logical View



The PT-LAPTOP-NM-1W module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop7

Physical

Config

Desktop

Programming

Attributes

MODULES

WPC300N

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

PT-HEADPHONE

PT-MICROPHONE

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View



Customize Icon in Logical View



The PT-LAPTOP-NM-1W module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop8

Physical

Config

Desktop

Programming

Attributes

MODULES

WPC300N

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

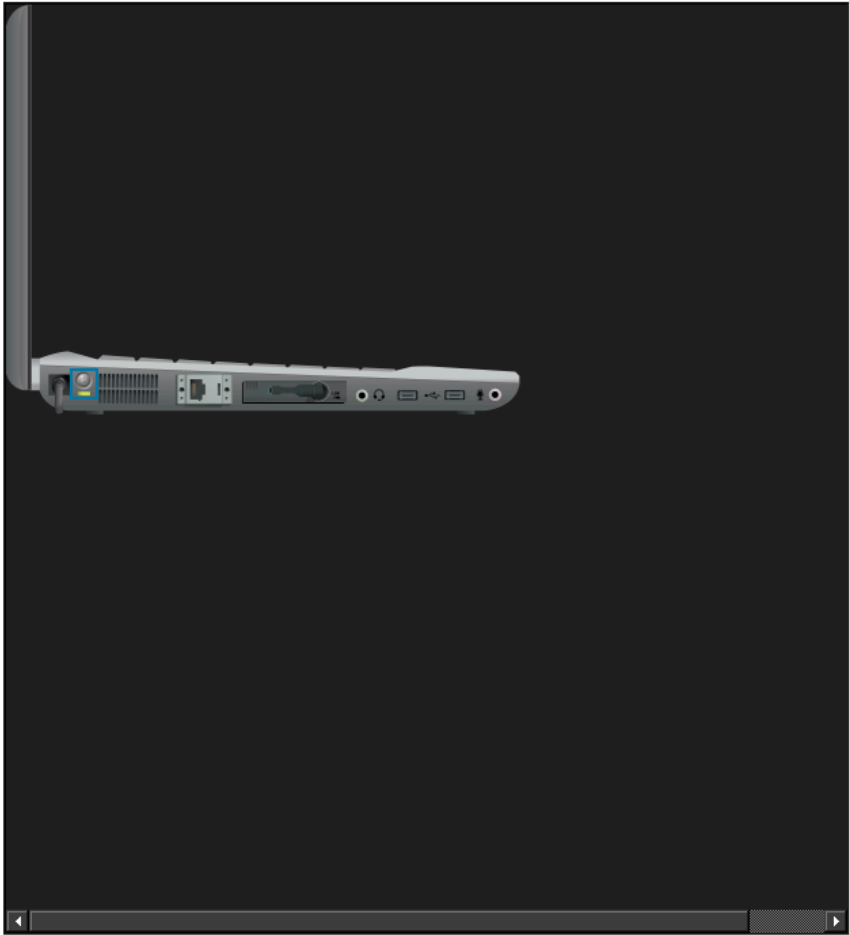
PT-HEADPHONE

PT-MICROPHONE

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View

Customize Icon in Logical View

The PT-LAPTOP-NM-1W module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop9

Physical

Config

Desktop

Programming

Attributes

MODULES

WPC300N

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

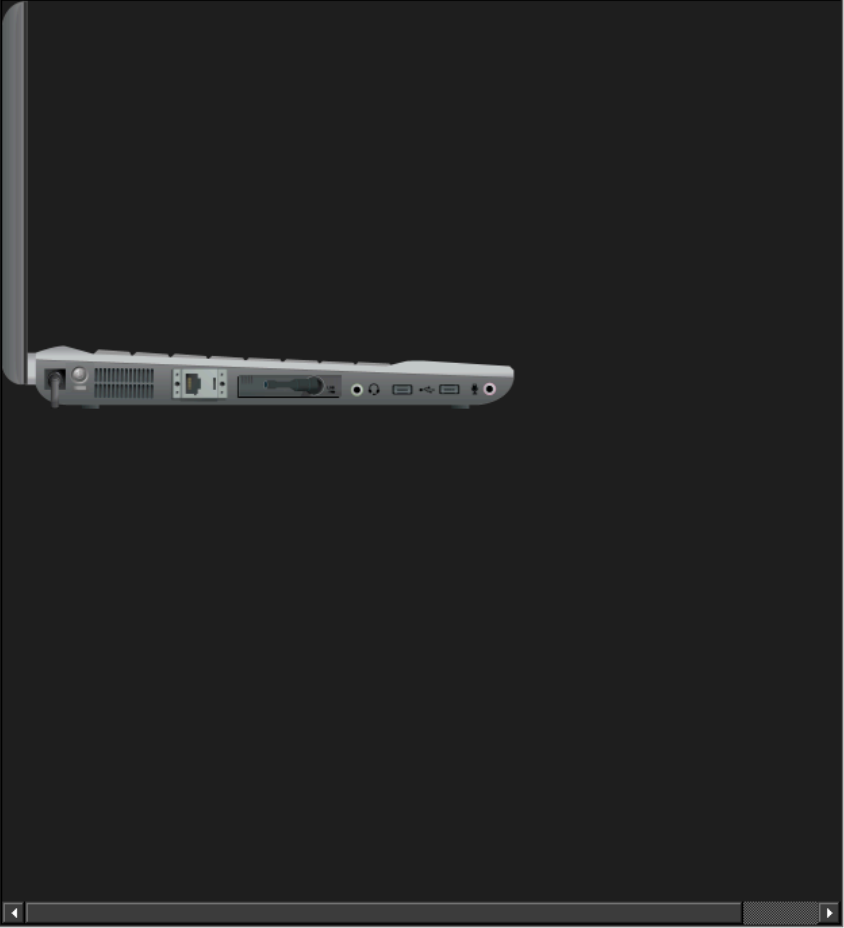
PT-HEADPHONE

PT-MICROPHONE

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View

Customize Icon in Logical View

The PT-LAPTOP-NM-1W module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop10

Physical

Config

Desktop

Programming

Attributes

MODULES

WPC300N

PT-LAPTOP-NM-1AM

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

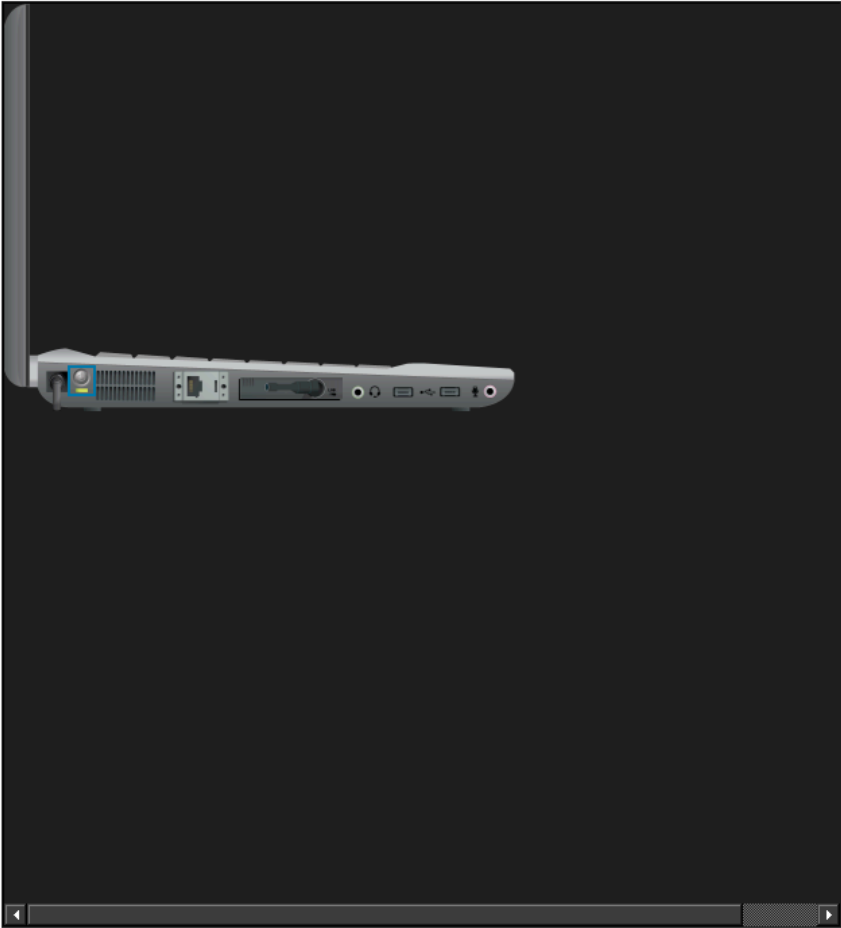
PT-HEADPHONE

PT-MICROPHONE

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View

Customize Icon in Logical View

The PT-LAPTOP-NM-1W module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.



Top

Laptop1

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status

On

Bandwidth

54 Mbps

MAC Address

0030.F216.A0A0

SSID

Wifi_Meeting

Authentication

Disabled

WPA-PSK

WPA

802.1X

WEP

WPA2-PSK

WPA2

Method:

WEP Key

PSK Pass Phrase

User ID

Password

User Name

Password

MD5

Encryption Type

AES

IP Configuration

DHCP

Static

IPv4 Address

192.168.77.16

Subnet Mask

255.255.255.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address

FE80::230:F2FF:FE16:A0A0

Laptop2

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status

☒ On

Bandwidth

54 Mbps

MAC Address

0030.A314.077D

SSID

Wifi_Meeting

Authentication

☒ Disabled

☒ WPA-PSK

☐ WPA

☒ 802.1X

☒ WEP

☒ WPA2-PSK

☐ WPA2

Method:

WEK Key

PSK Pass Phrase

User ID

Password

User Name

Password

SandiAman456

Encryption Type

AES

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

192.168.77.17

Subnet Mask

255.255.255.0

IPv6 Configuration

☒ Automatic

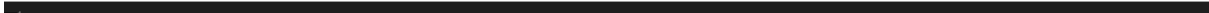
☐ Static

IPv6 Address

Link Local Address:

FE80::230:A3FF:FE14:77D

Top



Laptop4

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

Bluetooth

Global Settings

Display NameLaptop4

InterfacesWireless0

Gateway/DNS IPv4

DHCP

Static

Default Gateway192.168.77.1

DNS Server8.8.8.8

Gateway/DNS IPv6

Automatic

Static

Default Gateway

DNS Server

Device Clock:01:08:04 Mon Mar 1 1993 UTC

NTP

Server Ip:

Authentication

Enable

Disable

Password:

PTP

ProfilePower

Enable

Disable

Top

Laptop5

Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status

☒

On

Bandwidth

54 Mbps

MAC Address

0004.9A4A.2607

SSID

Wifi_Meeting

Authentication

☒ Disabled

☒ WPA-PSK

☒ WPA

☒ 802.1X

☒ WEP

☒ WPA2-PSK

☒ WPA2

Method:

WEK Key

PSK Pass Phrase

User ID

Password

User Name

Password

SandiAman456

Encryption Type

AES

IP Configuration

☒ DHCP

☒ Static

IPv4 Address

192.168.77.22

Subnet Mask

255.255.255.0

IPv6 Configuration

☒ Automatic

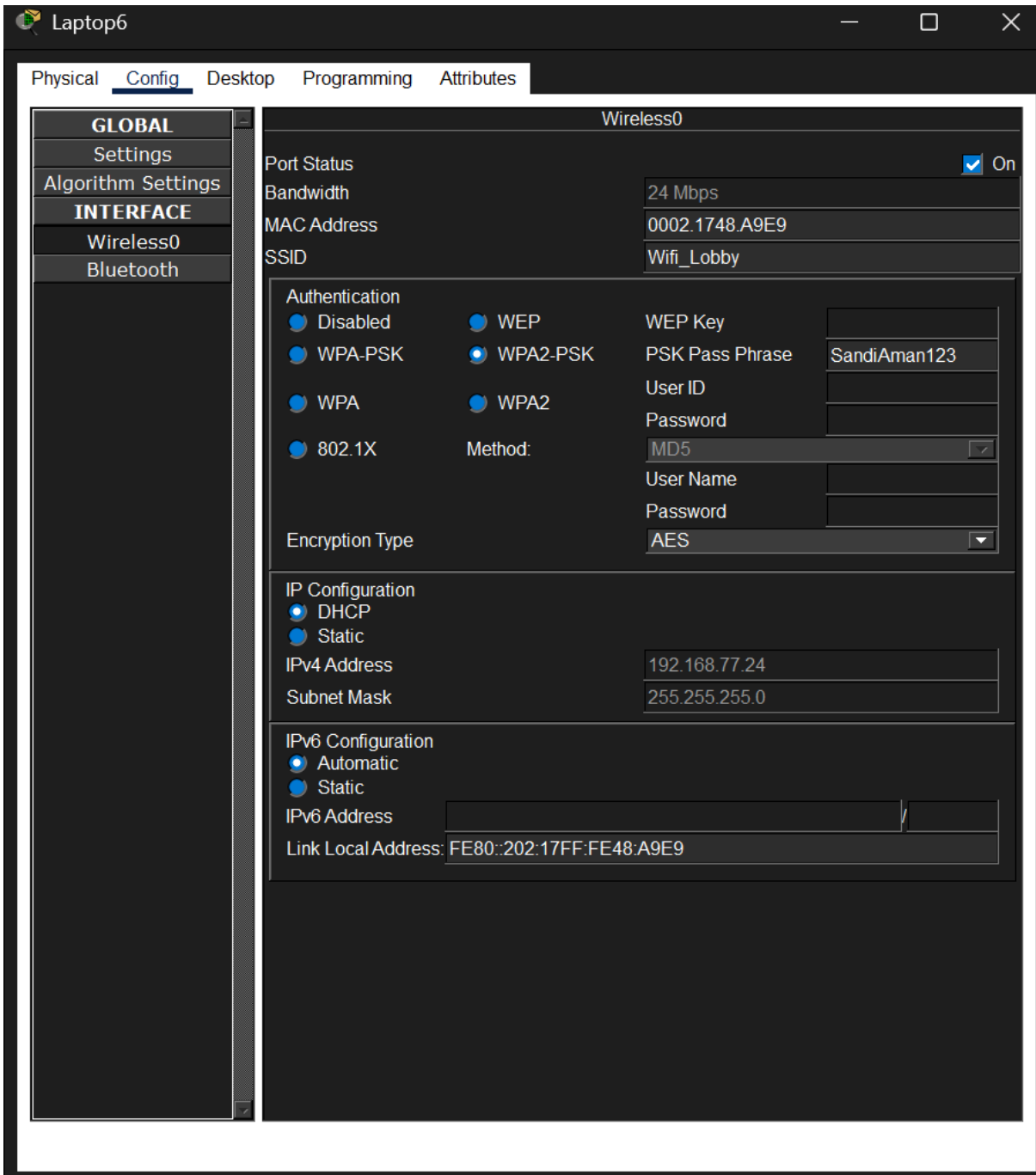
☒ Static

IPv6 Address

Link Local Address

FE80::204:9AFF:FE4A:2607

Top



Laptop8

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status

On

Bandwidth

24 Mbps

MAC Address

0060.3E54.50AB

SSID

Wifi_Lobby

Authentication

Disabled

WPA-PSK

WPA

802.1X

WEP

WPA2-PSK

WPA2

Method:

WEP Key

PSK Pass Phrase

User ID

Password

User Name

Password

MD5

Encryption Type

AES

IP Configuration

DHCP

Static

IPv4 Address

Subnet Mask

192.168.77.27

255.255.255.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local Address:

FE80::260:3EFF:FE54:50AB

Top

GLOBAL		Wireless0	
Settings		Port Status ☒ On	
Algorithm Settings		Bandwidth 18 Mbps	
INTERFACE		MAC Address 0040.0B2C.97E1	
Wireless0		SSID Wifi_Lobby	
Bluetooth			
Authentication ☒ Disabled ☐ WEP WEP Key ☐ WPA-PSK ☒ WPA2-PSK PSK Pass Phrase SandiAman123 ☐ WPA ☐ WPA2 User ID ☒ 802.1X Method: MD5 Password			
		Encryption Type AES	
IP Configuration ☒ DHCP ☐ Static IPv4 Address 192.168.77.28 Subnet Mask 255.255.255.0			
IPv6 Configuration ☒ Automatic ☐ Static IPv6 Address Link Local Address: FE80::240:BFF:FE2C:97E1			

Laptop10

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status

On

Bandwidth18 Mbps

MAC Address000C.858B.2995

SSIDWifi_Lobby

Authentication

Disabled

WPA-PSK

WPA

802.1X

WEP

WPA2-PSK

WPA2

Method:

WEP Key

PSK Pass PhraseSandiAman123

User ID

Password

User Name

Password

Encryption TypeAES

IP Configuration

DHCP

Static

IPv4 Address192.168.77.30

Subnet Mask255.255.255.0

IPv6 Configuration

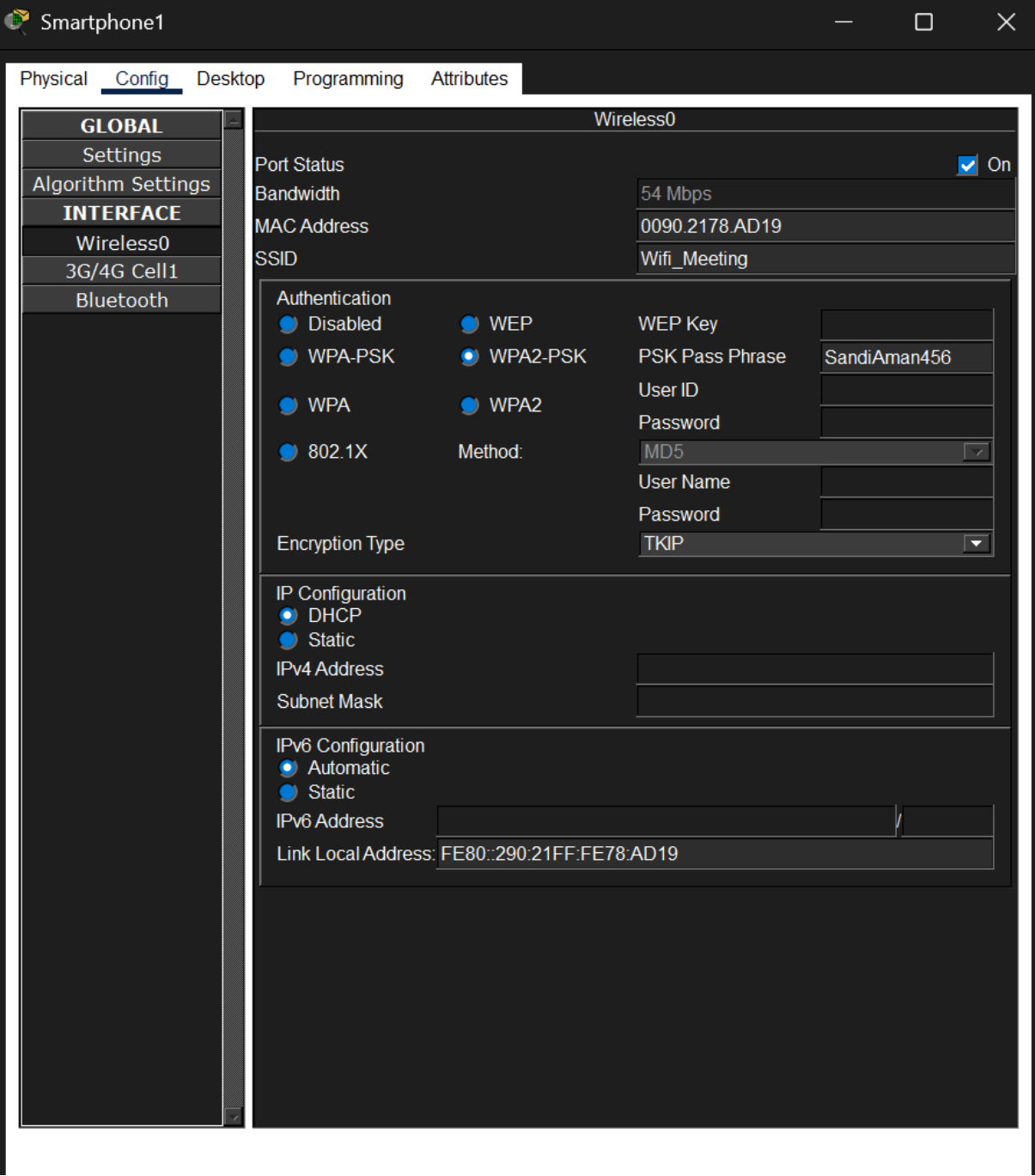
Automatic

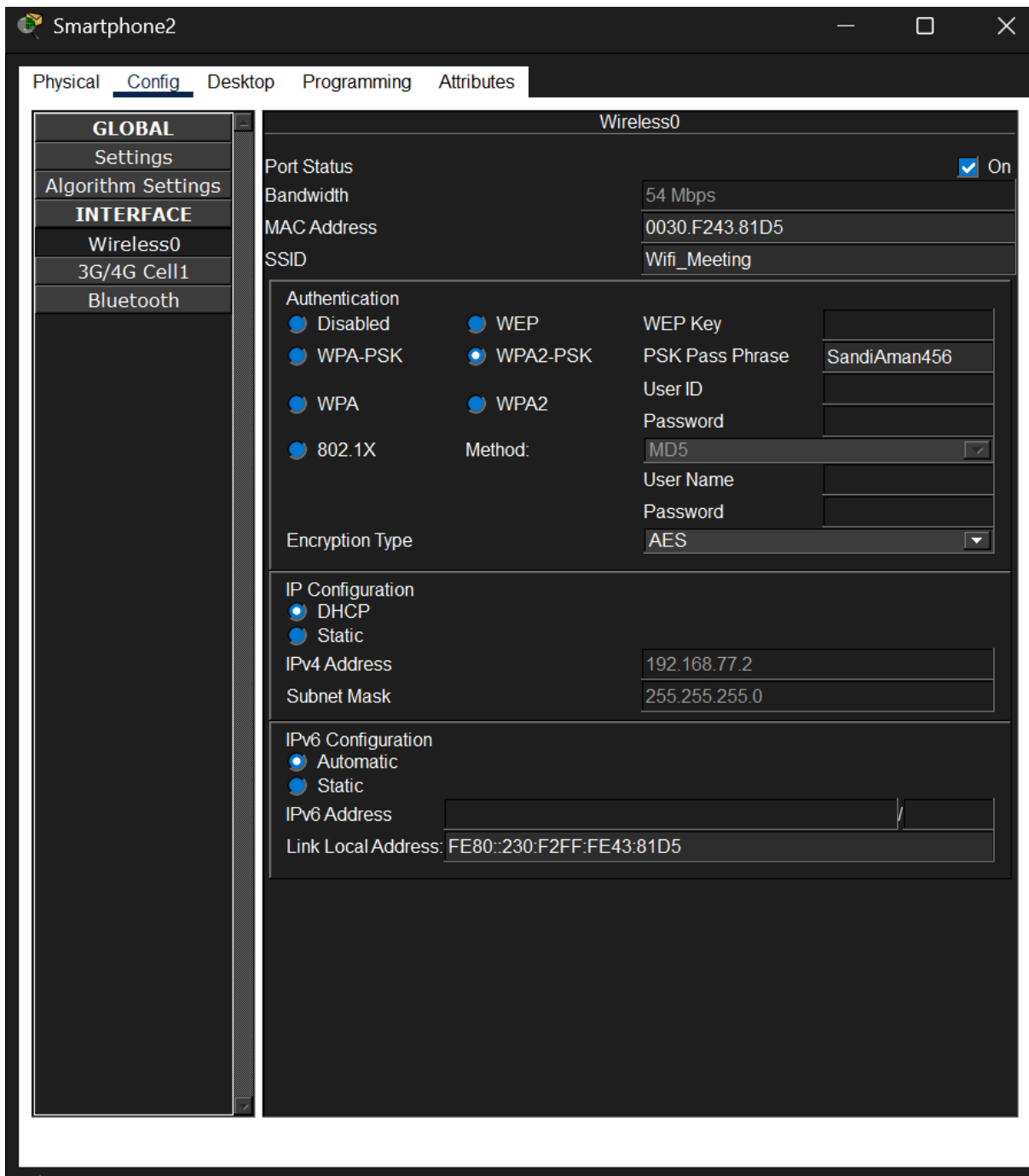
Static

IPv6 Address

Link Local AddressFE80::20C:85FF:FE8B:2995

Top





Smartphone3

Physical
Config
Desktop
Programming
Attributes

GLOBAL
Settings
Algorithm Settings
INTERFACE
Wireless0
3G/4G Cell1
Bluetooth

Wireless0

Port Status

☒
On

Bandwidth

54 Mbps

MAC Address

00E0.A377.27A1

SSID

Wifi_Meeting

Authentication

☒ Disabled
☒ WEP
☒ WPA-PSK
☒ WPA2-PSK
☒ WPA
☒ WPA2
☒ 802.1X

Method:

MD5

WEP Key

PSK Pass Phrase

SandiAman456

User ID

Password

User Name

Password

Encryption Type

AES

IP Configuration

☒ DHCP
☒ Static

IPv4 Address

192.168.77.4

Subnet Mask

255.255.255.0

IPv6 Configuration

☒ Automatic
☒ Static

IPv6 Address

Link Local Address:

FE80::2E0:A3FF:FE77:27A1

The wireless settings for smartphones 4–10 are the same as for 1–3, except that starting with wireless 6, the SSID is changed to Wifi_Lobby and the password is changed to SandiAman123

PC1

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☒ Static

IPv4 Address192.168.77.31

Subnet Mask255.255.255.0

Default Gateway192.168.77.1

DNS Server8.8.8.8

IPv6 Configuration

☒ Automatic

☒ Static

IPv6 Address/

Link Local AddressFE80::2E0:F7FF:FE90:233D

Default Gateway

DNS Server

802.1X

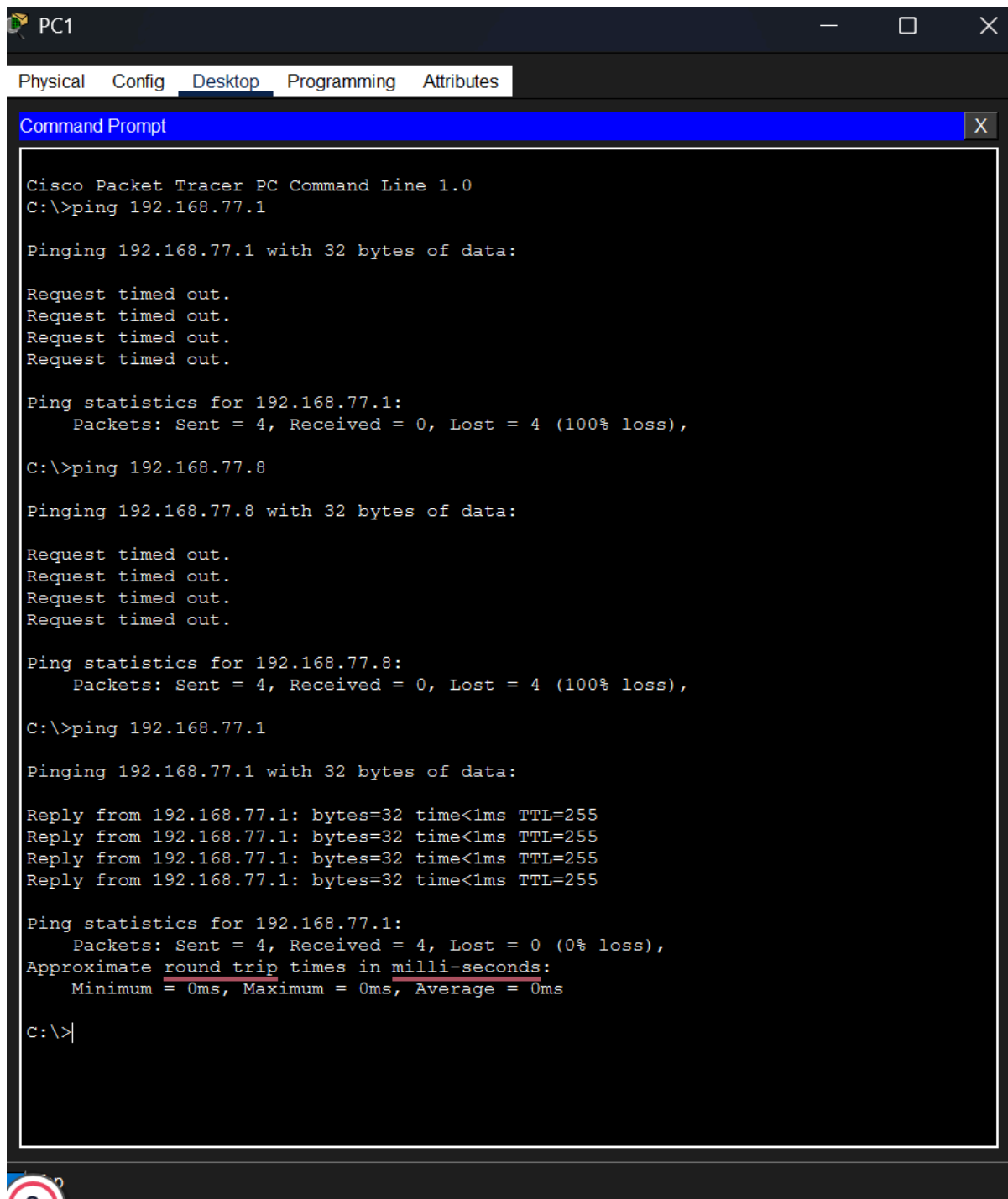
☐ Use 802.1X Security

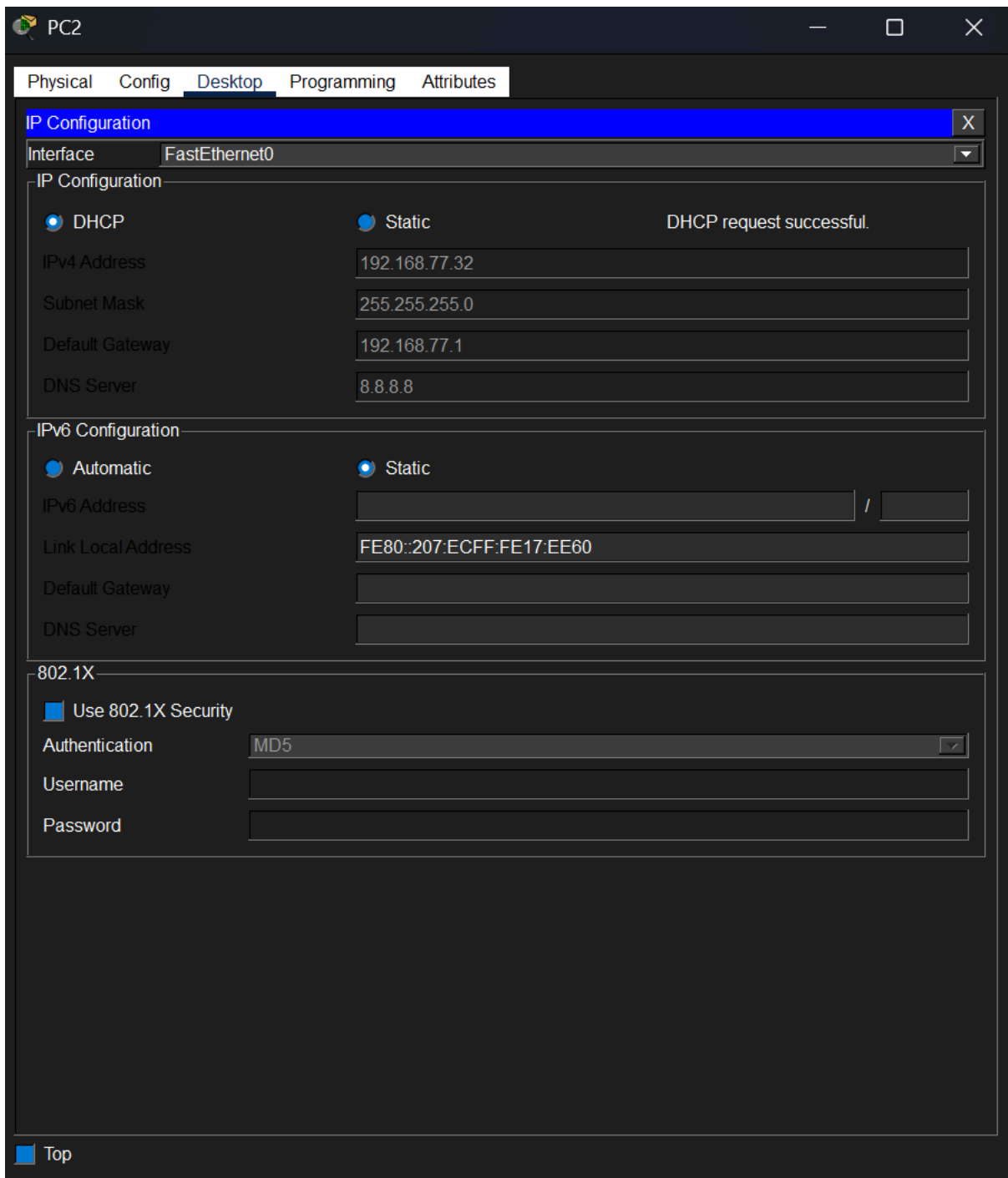
AuthenticationMD5

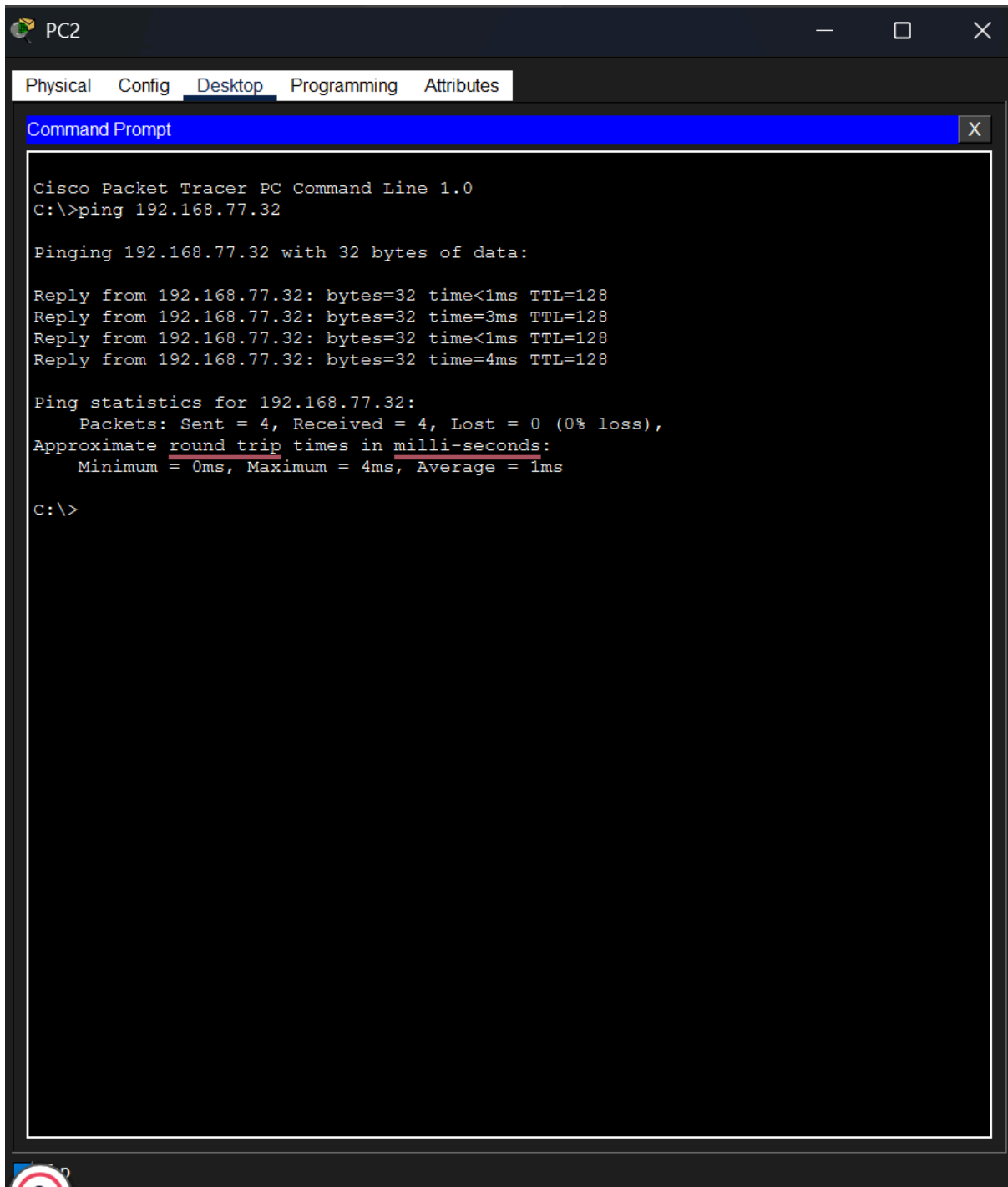
Username

Password

Top







The screenshot shows a Cisco Packet Tracer interface with a window titled "PC2". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the execution of a ping command to 192.168.77.32, with successful results and statistics.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.77.32

Pinging 192.168.77.32 with 32 bytes of data:

Reply from 192.168.77.32: bytes=32 time<1ms TTL=128
Reply from 192.168.77.32: bytes=32 time=3ms TTL=128
Reply from 192.168.77.32: bytes=32 time<1ms TTL=128
Reply from 192.168.77.32: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.77.32:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>
```

Follow the same steps for PCs 3 through 5

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Request timed out.

Ping statistics for 192.168.77.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.77.8

Pinging 192.168.77.8 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.77.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.77.1

Pinging 192.168.77.1 with 32 bytes of data:

Reply from 192.168.77.1: bytes=32 time<1ms TTL=255
Reply from 192.168.77.1: bytes=32 time<1ms TTL=255
Reply from 192.168.77.1: bytes=32 time<1ms TTL=255
Reply from 192.168.77.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.77.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.77.16

Pinging 192.168.77.16 with 32 bytes of data:

Reply from 192.168.77.16: bytes=32 time=18ms TTL=128
Reply from 192.168.77.16: bytes=32 time=48ms TTL=128
Reply from 192.168.77.16: bytes=32 time=40ms TTL=128
Reply from 192.168.77.16: bytes=32 time=40ms TTL=128

Ping statistics for 192.168.77.16:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 48ms, Average = 36ms

C:\>
```

